THE MAXIMAL WEAKLY ETALE SUBALGEBRA COMMUTES WITH AN EXTENSION OF THE BASE FIELD

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Allowed standard prerequisites (editorial).

- The maximal weakly etale subalgebra of an algebra over a field is weakly etale, is preserved by base change of the field, and is a field when the algebra is reduced and geometrically irreducible. invoked at proof lines 30988-30994 and again at proof lines 31064-31072. Identifies the two sides of the comparison map and supplies the field structure used in the final computation.
- Base change of a weakly etale algebra along a field extension is weakly etale, which makes the comparison map injective. invoked at proof lines 30991-30993. Produces the canonical injective map from the base changed subalgebra into the maximal subalgebra of the base changed algebra.
- The maximal weakly etale subalgebra is compatible with filtered colimits, so a statement about finite type algebras implies the general case. invoked at proof lines 31003-31006. Reduces the proof to the case of a finite type algebra over the base field.
- Weakly etale algebras over a field are reduced, which lets the nilpotent kernel of the reduction be discarded. invoked at proof lines 31010-31014. Shows that the comparison map stays injective when the algebra is replaced by its reduced quotient.
- A finite separable extension of fields becomes a product of copies of the larger field after a base change to a separably closed or algebraically closed field. invoked at proof lines 31083-31088. Decomposes the tensor product of the algebra over the finite separable field with the algebraically closed field into a product of factors.

Source reference. Original lemma 092H: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30303–30303. Used at proof lines 30991-30993 to see that the base changed maximal subalgebra is weakly etale over the larger field, which produces the comparison map; everything built on that map is inside the counted span.

Source reference. Original lemma 0CKS: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30892–30892. The statement names the object through this lemma; it is a pointer fixing notation, and every property of that object used later is invoked explicitly and applied inside the counted span.

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Source reference. Original lemma 0CKT: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30913–30913. Invoked for compatibility with filtered colimits at proof lines 31003-31006, for the behaviour under reduction at proof lines 31013-31016 and for the field and geometric irreducibility properties at proof lines 31064-31072; each application and the computation that follows are inside the counted span.

Source reference. Original lemma 0CKR: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30846–30846. Used at proof lines 31010-31012 to conclude that the maximal subalgebra over the larger field is reduced, which is what makes the reduction step work; the reduction itself is inside the counted span.

Source reference. Original lemma 092L: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30405–30405. The final sentence at proof lines 31098-31101 uses it to name the map whose weak etaleness gives the equality; the decomposition producing that map is inside the counted span.

Source reference. Original lemma 00FR: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer for the decompositions of spectra into irreducible components used while reducing to the reduced case; the use made of the decomposition is inside the counted span.

Source reference. Original lemma 00FN: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer for the permanence statements the reduction to finite type algebras relies on; the reduction itself and its consequences are inside the counted span.

Source reference. Original lemma 037Q: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 31071-31072 to see that the algebra is geometrically irreducible over the finite separable field of constants; the decomposition that uses this is inside the counted span.

Source reference. Original lemma 00EU: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer used during the reduction to the reduced case; the comparison map and the reduction step are inside the counted span.

Source reference. Original lemma 02LX: sources/stacks-chow/tags.txt, source lines 1–1. A navigation pointer for the manipulation of the field of constants in the reduced case; the identification of that field is inside the counted span.

Source reference. Original lemma 0CKN: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 31085-31090 for the displayed isomorphism splitting the tensor product into copies of the algebraically closed field; the subsequent factor by factor computation of the maximal subalgebras is inside the counted span.

Lemma 0.2. *Let L/K be an extension of fields. Let A be a K-algebra. Let $B \subset A$ be the maximal weakly étale K-subalgebra of A as in Lemma 0CKS. Then $B \otimes_K L$ is the maximal weakly étale L-subalgebra of $A \otimes_K L$.*

Proof. For an algebra A over K we write $B_{max}(A/K)$ for the maximal weakly étale K-subalgebra of A. Similarly we write $B_{max}(A'/L)$ for the maximal weakly étale L-subalgebra of A' if A' is an L-algebra. Since $B_{max}(A/K) \otimes_K L$ is weakly étale over L (Lemma 092H) and since $B_{max}(A/K) \otimes_K L \subset A \otimes_K L$ we obtain a canonical injective map

$$B_{max}(A/K) \otimes_K L \to B_{max}((A \otimes_K L)/L)$$

The lemma states that this map is an isomorphism.

To prove the lemma for L and our K-algebra A, it suffices to prove the lemma for any field extension L' of L. Namely, we have the factorization

$$B_{max}(A/K) \otimes_K L' \to B_{max}((A \otimes_K L)/L) \otimes_L L' \to B_{max}((A \otimes_K L')/L')$$

hence the composition cannot be surjective without $B_{max}(A/K) \otimes_K L \to B_{max}((A \otimes_K L)/L)$ being surjective. Thus we may assume L is algebraically closed.

Reduction to finite type K-algebra. We may write A is the filtered colimit of its finite type K-subalgebras. Using Lemma 0CKT we see that it suffices to prove the lemma for finite type K-algebras.

Assume A is a finite type K-algebra. Since the kernel of $A \to A_{red}$ is nilpotent, the same is true for $A \otimes_K L \to A_{red} \otimes_K L$. Then

$$B_{max}((A \otimes_K L)/L) \to B_{max}((A_{red} \otimes_K L)/L)$$

is injective because the kernel is nilpotent and the weakly étale L-algebra $B_{max}((A \otimes_K L)/L)$ is reduced (Lemma 0CKR). Since $B_{max}(A/K) = B_{max}(A_{red}/K)$ by Lemma 0CKT we conclude that it suffices to prove the lemma for A_{red}.

Assume A is a reduced finite type K-algebra. Let $Q = Q(A)$ be the total quotient ring of A. Then $A \subset Q$ and $A \otimes_K L \subset Q \otimes_A L$ and hence

$$B_{max}(A/K) = A \cap B_{max}(Q/K)$$

and

$$B_{max}((A \otimes_K L)/L) = (A \otimes_K L) \cap B_{max}((Q \otimes_K L)/L)$$

by Lemma 0CKT. Since $- \otimes_K L$ is an exact functor, it follows that if we prove the result for Q, then the result follows for A. Since Q is a finite product of fields (Algebra, Lemmas 02LX, 00EU, 00FR, and 00FN) and since B_{max} commutes with products (Lemma 0CKT) it suffices to prove the lemma when A is a field.

Assume A is a field. We reduce to A being finitely generated over K by the argument in the third paragraph of the proof. (In fact the way we reduced to the case of a field produces a finitely generated field extension of K.)

Assume A is a finitely generated field extension of K. Then $K' = B_{max}(A/K)$ is a field separable algebraic over K by Lemma 0CKT part (6). Hence K' is a finite separable field extension of K and A is geometrically irreducible over K' by Algebra, Lemma 037Q. Since L is algebraically closed and K'/K finite separable we see that

$$K' \otimes_K L \to \prod\nolimits_{\sigma \in \mathrm{Hom}_K(K', L)} L, \quad \alpha \otimes \beta \mapsto (\sigma(\alpha)\beta)_\sigma$$

is an isomorphism (Fields, Lemma 0CKN). We conclude

$$A \otimes_K L = A \otimes_{K'} (K' \otimes_K L) = \prod\nolimits_{\sigma \in \mathrm{Hom}_K(K', L)} A \otimes_{K', \sigma} L$$

Since A is geometrically irreducible over K' we see that $A \otimes_{K', \sigma} L$ has a unique minimal prime. Since L is algebraically closed it follows that $B_{max}((A \otimes_{K', \sigma} L)/L) = L$ because this L-algebra is a field algebraic over L by Lemma 0CKT part (6). It follows that the maximal weakly étale $K' \otimes_K L$-subalgebra of $A \otimes_K L$ is $K' \otimes_K L$ because we can decompose these subalgebras into products as above. Hence the inclusion $K' \otimes_K L \subset B_{max}((A \otimes_K L)/L)$ is an equality: the ring map $K' \otimes_K L \to B_{max}((A \otimes_K L)/L)$ is weakly étale by Lemma 092L. □